

École Pour l'Informatique et les Techniques Avancées – EPITA

Masters program

Course: Introduction to Network protocols and Architecture (INPA)

INPA Course Outline

Course schedule (tentative)

Date	No.	Topics	Duration (in hours)
(check Zeus platform for the date & time)	1	Primer, TCP/IP stack, TCP/IP stack operations, Class exercises	3
(check Zeus platform for the date & time)	2	IP address allocation & routing, Subnetting, ARP, Class exercises	3
(check Zeus platform for the date & time)	3	DHCP, Packet handling, Routing Protocols (RIP, OSPF), Class exercises	3
(check Zeus platform for the date & time)	4	TCP, UDP, ICMP, ON, Class exercise	3
(check Zeus platform for the date & time)	5	Internet layer protocols (IGP vs EGP, BGP), ACLs, Class exercises	3
(check Zeus platform for the date & time)	6	Application Layer protocols/Security (HTTP/S), Class exercise	3
(check Zeus platform for the date & time)	7	Application Layer protocols/Security (SMTP + PGP), DNS, Class exercises	3
(check Zeus platform for the date & time)	8	NAT, Class exercise, Review/Practice-session	3
Total			24 hours
(check Zeus platform for the date & time)	EXAM		1

GRADING criteria:

Class attendance and responsiveness:	10%
Class assignments (practical work):	40%
Final evaluation (Quiz & Exam):	50%

Lecture 3 Outline

- ▶ **DHCP**
 - DHCP anatomy
 - Messages & operation
 - DHCP configuration
 - Class exercise 7
- ▶ **Packet handling**
 - Circuit switching vs packet switching
 - Packet switching approaches
 - Switching types comparison
- ▶ **Routing Information protocol (RIP)**
 - RIP operation
 - RIP v1 & v2
 - Class exercise 8
- ▶ **Open Shortest Path First (OSPF)**
 - Operations
 - Shortest Path Algorithm (Dijkstra)
 - Router types
 - Subdivided areas
 - Class exercise 9

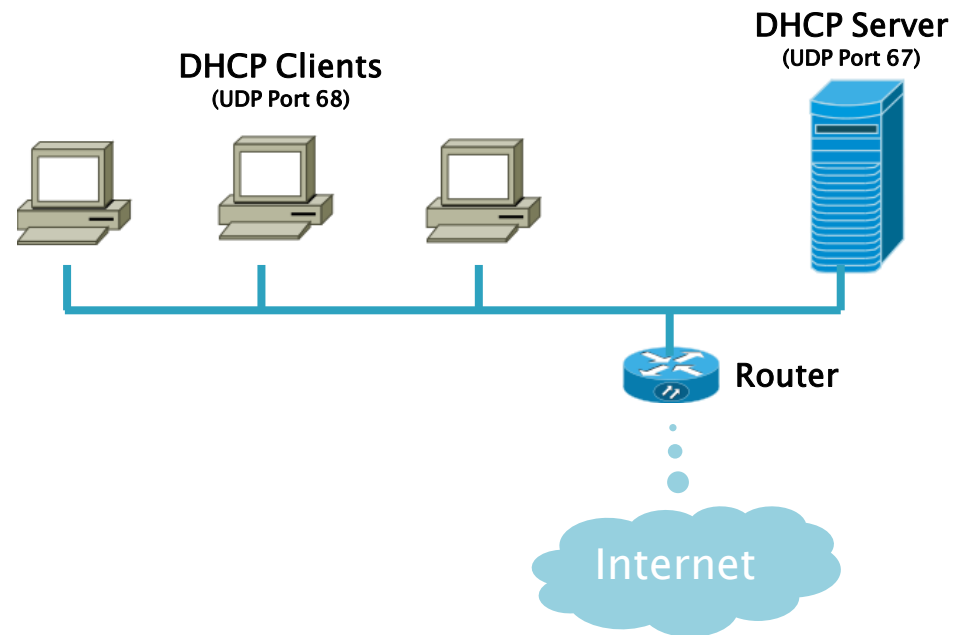
DHCP protocol & motivations

- ▶ DHCP (Dynamic Host Configuration Protocol) allows dynamic allocation of IP addresses
 - Minimal human intervention
 - Standardized under RFC1541 and RFC2131 (current)
- ▶ Configuration parameters for network hosts
 - IP address
 - Router
 - Subnet Mask
 - ...
- ▶ Before DHCP
 - Manuel assignment
 - Obsolete/superseded protocols (e.g., RARP, BOOTP)

DHCP Anatomy

- ▶ Uses the same IANA assigned ports as BOOTP
 - 67/udp for the server (S), 68/udp for the client (C)

1. C: Discover
2. S: Offer
3. C: Request
4. S: Acknowledge
5. C: Inform
6. S: Release

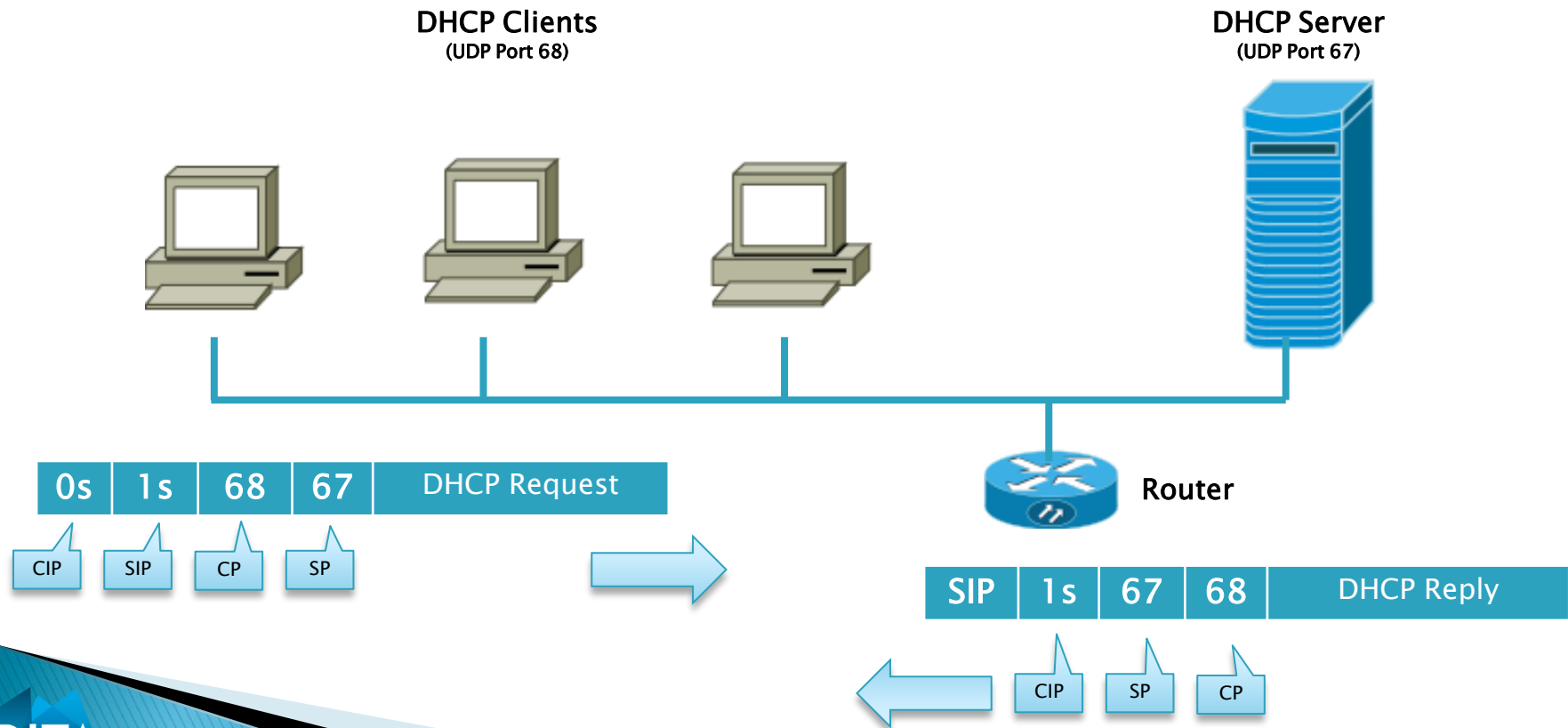


Leases

- ▶ Server offers IP address and network parameters for a limited time (called a lease)
 - In practice, leases may vary from 30 minutes to a week or so
- ▶ Short lease:
 - Clients get updated parameters quickly
 - Essential if you have more clients than addresses
- ▶ Long lease:
 - More reliable (clients may continue to operate for a week after DHCP server fails)

Client and server on the same network

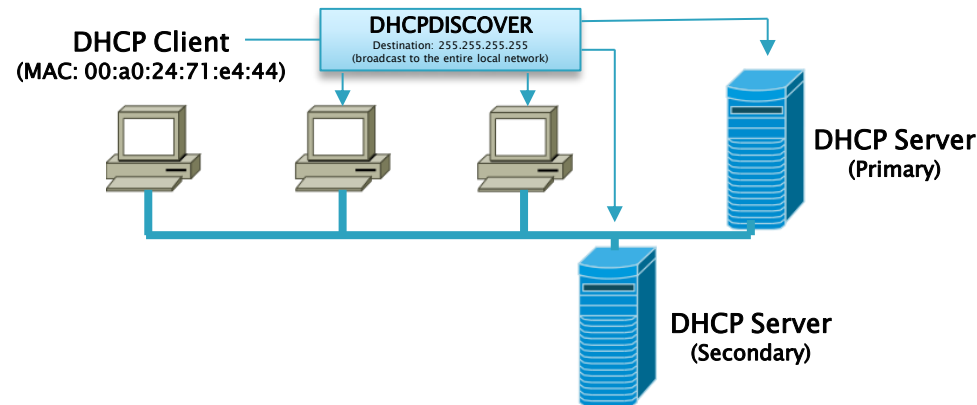
CIP: Client IP Address, **CP:** Client Port Number
SIP: Server IP Address, **SP:** Server Port Number



DHCP Operation

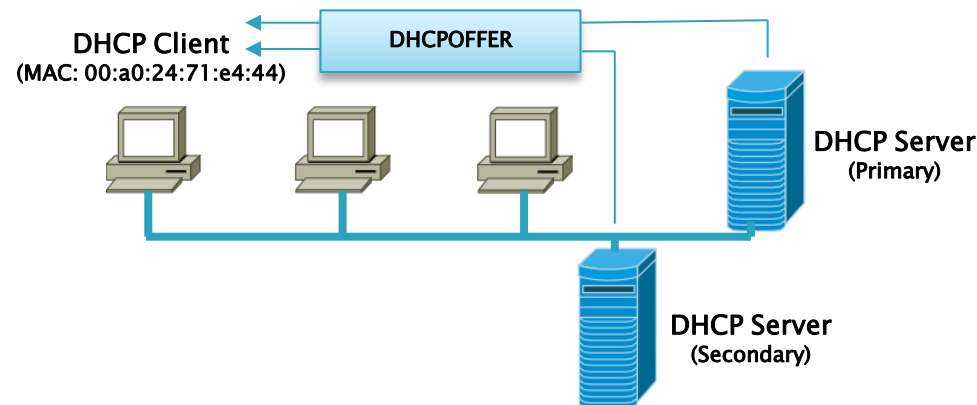
1: DHCP DISCOVER (broadcast)

→ IP address lease request



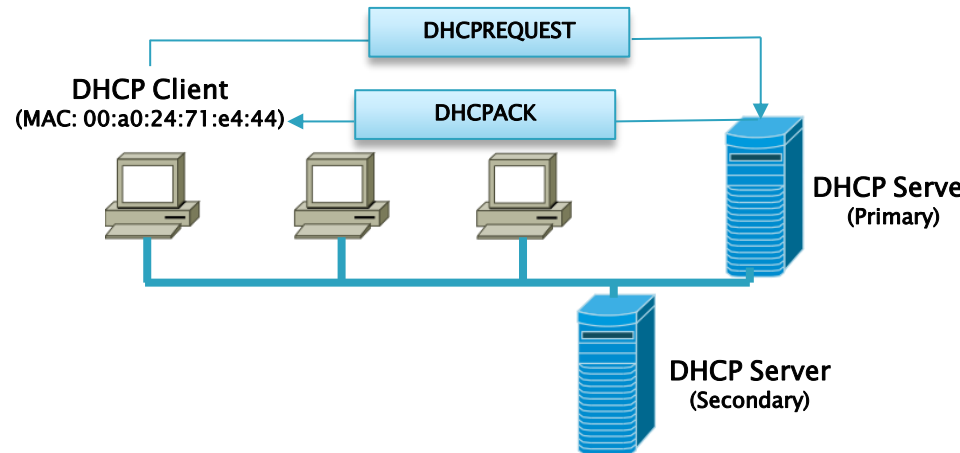
2: DHCP OFFER

→ DHCP server reserves an IP address for the client and makes a lease offer

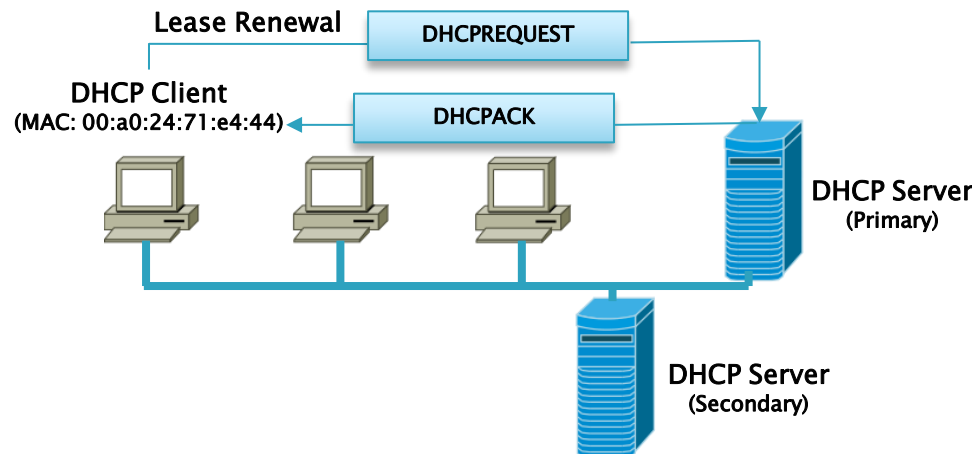


DHCP Operation (cont.)

3: DHCP client sends DHCPREQUEST (requesting the offered IP address). The server, then sends DHCPACK
→ At this time, the DHCP client can start using the IP address



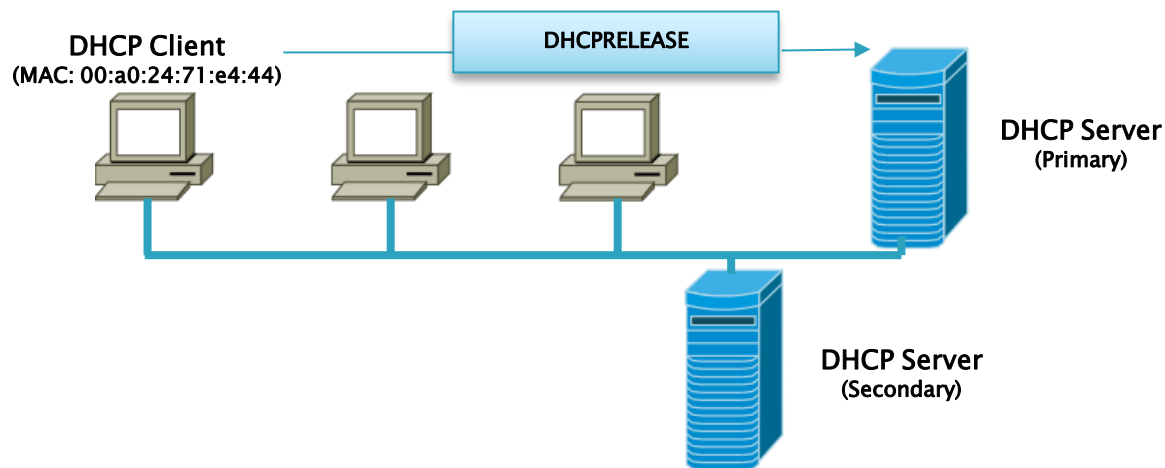
4: Renewing a Lease (sent when 50% of lease has expired)



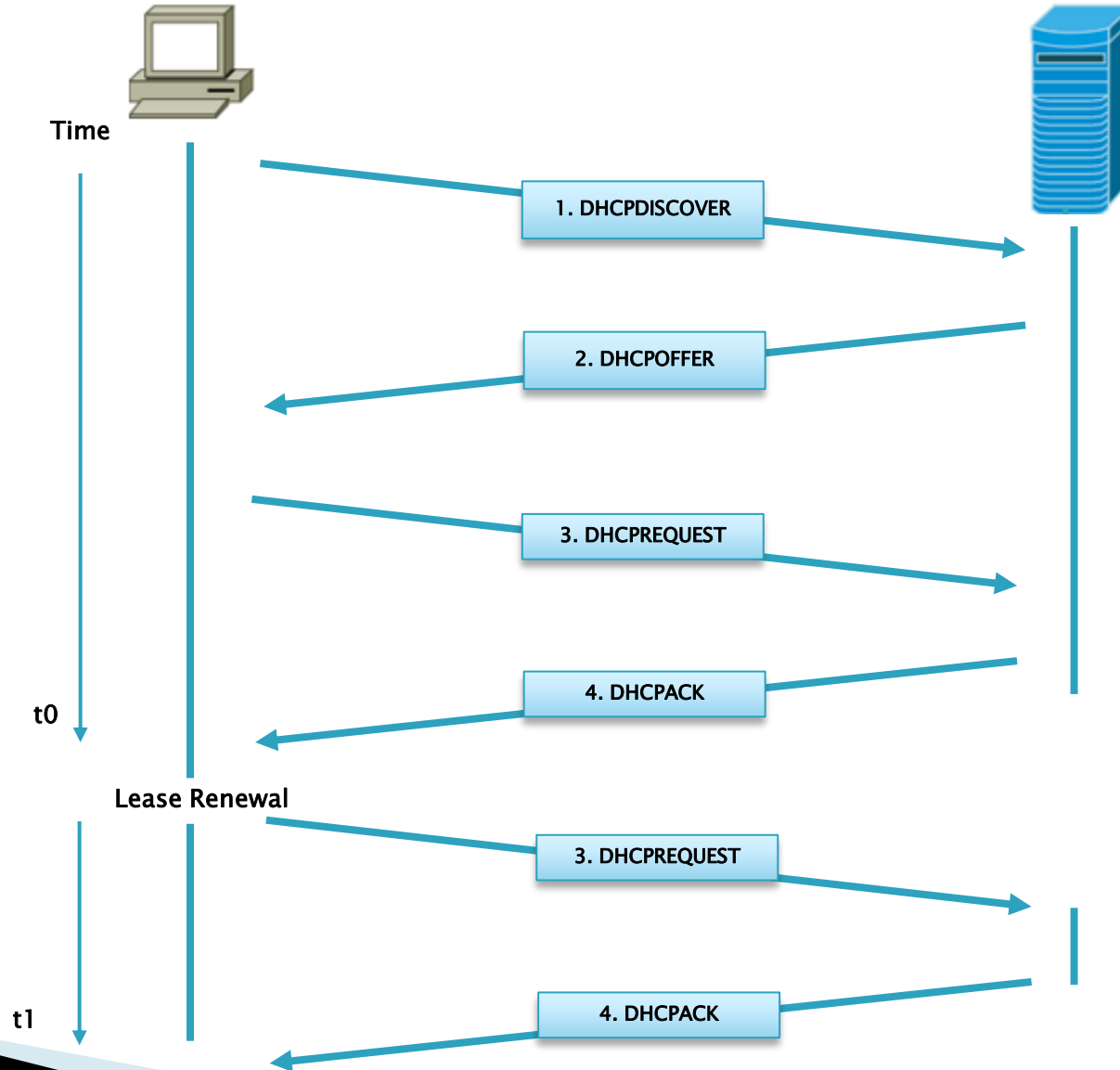
DHCP Operation (cont.)

5: DHCP RELEASE

→ Client deactivates its IP address by sending DHCP RELEASE. As client devices usually do not know when users may unplug them from the network, the protocol does not mandate the sending of *DHCP Release*



DHCP messages (visually)



/etc/dhcp/dhcpd.conf

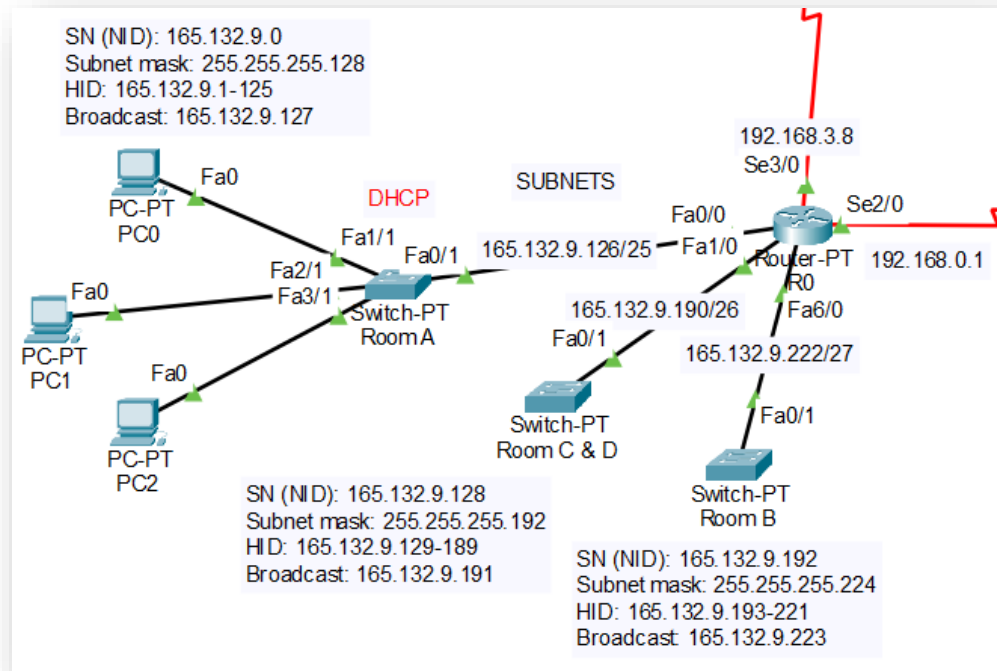
```
default-lease-time 600;  
max-lease-time 7200;
```

```
subnet 192.168.1.0 netmask 255.255.255.0 {  
  range 192.168.1.10 192.168.1.100;  
  option subnet-mask 255.255.255.0;  
  option broadcast-address 192.168.1.255;  
  option routers 192.168.1.254;  
  option domain-name-servers 192.168.1.1,  
  192.168.1.2;  
  option domain-name "ubuntu-fr.lan";  
}
```

Exercise 7: Practical work

Use your existing cisco packet tracer file (from last class exercise):

1. On R0: Set up a DHCP server:
 1. Config mode:
 - *ip dhcp pool POOLNAME*
 - Define network:
*network 192.168.x.x
255.255.255.X*
 - Define gateway:
default-router 192.168.X.X
 - Exclude some IP address from the pool:
*ip dhcp excluded-address
X.X.X.X Z.Z.Z.Z*
 2. Add PC0, PC1 and PC2 via Room A switch to R0
 - Use DHCP for IP assignments



Save cisco packet tracer file with your 'First_Last-name_Ex-3-7'

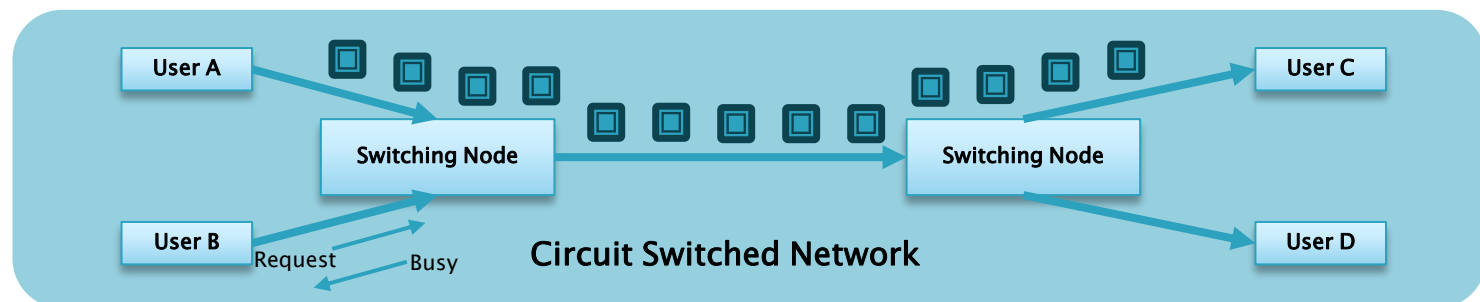
Deadline: See 'Teams' Assignment section

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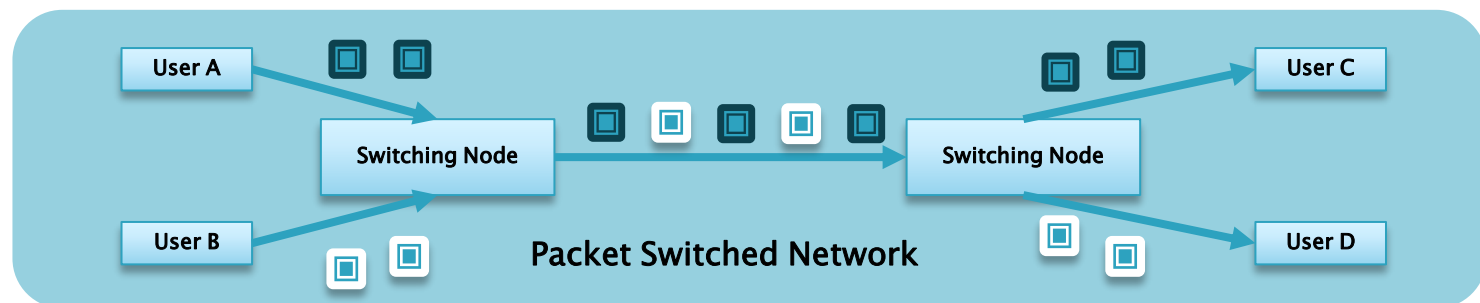
Circuit Switching

- ▶ In Circuit Switching (CS), network dedicated channel must be established before the call is made between users
 - The channel is then reserved (with dedicated transmission rate) till the connection is active e.g., phone call over PSTN (Public switched telephone network), POTS (plain old telephone service)
 - Any idle resource that is part of the reserved channel, then cannot be used by other ongoing communications



Packet Switching

- ▶ In packet switching network, unlike CS network, it is not required to establish the connection initially
 - The connection/channel is available to use by many users and allocates variable transmission rate based on demand
 - Message is broken into series of small packets (that contains their routing information)



Packet switching approaches

- ▶ Packet switching can be viewed under two approaches:
 - **Datagram type (connection-less):** e.g., IP, UDP,
 - Each packet (containing source/destination IP, Port, sequence) is treated independently
 - Every intermediate node must take routing decisions
 - **Virtual circuit type (connection-oriented):** e.g., TCP
 - A preplanned route is established before any packets are sent
 - Route is fixed for the duration of logical connection for all packets
 - For route termination, a clear request must be sent

Virtual circuit can be regarded as 'mixed' (i.e., share some properties of circuit switching)

Comparing switching types

Circuit Switching	Packet Switching (Datagram type)	Packet Switching (Virtual Circuit type)
Dedicated path	No Dedicated path	No Dedicated path
Path is established for entire conversation	Route is established for each packet	Route is established for entire conversation
Call setup delay	Packet transmission delay	Call setup delay as well as packet transmission delay
Overload may block call setup	Overload increases packet delay	Overload may block call setup and increases packet delay
Fixed bandwidth	Dynamic bandwidth	Dynamic bandwidth
No overhead bits after call setup	Overhead bits in each packet	Overhead bits in each packet

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Routing protocols

	Interior Gateway Protocols				Exterior Gateway Protocols
	Distance Vector Routing Protocols		Link State Routing Protocols		Path Vector
Classful	RIP	IGRP			EGP
Classless	RIPv2	EIGRP	OSPFv2	IS-IS	BGPv4
IPv6	RIPng	EIGRP for IPv6	OSPFv3	IS-IS for IPv6	BGPv4 for IPv6

- ▶ **RIPv1 (Routing Information Protocol):**
Distance Vector, Classful Routing Protocol
 - RIP use UDP on port 520
 - Uses hop count (i.e., Distance) as its only metric for path selection; Vector: direction

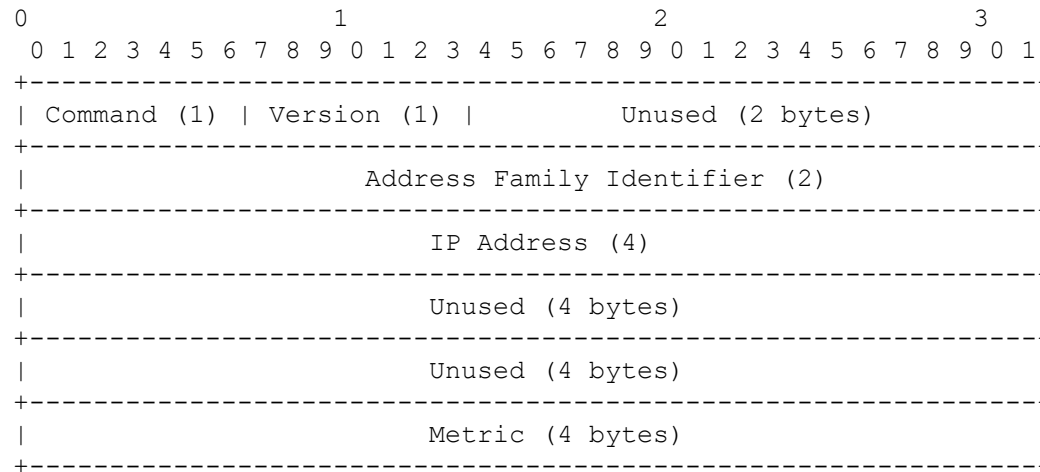
Routing Information Protocol (RIP)

- ▶ RIP evolved from the Xerox Network System (NS) in the late 1970's
 - In 1988, it was standardized under RFC 1058
- ▶ Why learn RIP?
 - Still in use today (mostly V2)
 - Help understand fundamental concepts and comparisons of protocols such as classful (RIPv1) and classless (RIPv2)
 - IPv6 form of RIP: RIPng (next generation)

RIP Operation

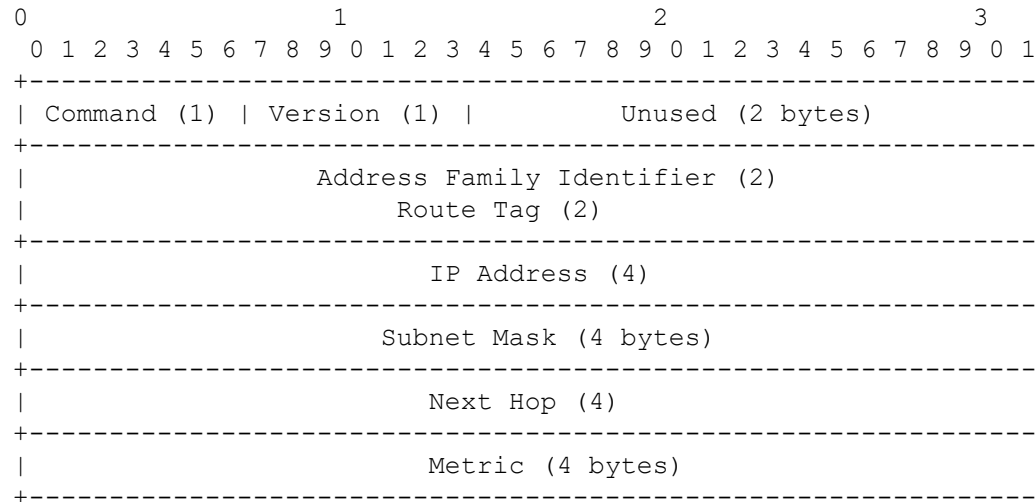
1. On Start-up:
 - Each RIP-configured interface broadcasts a request message, asking any RIP neighbors to send their complete routing table
2. Each RIP neighbor responds with the information
3. The requesting router evaluates each route
 - If it's a new route, it gets added to the routing table
 - If it's already in the routing table and has a better hop count (lower), the routing table is updated
 - If there are no changes, it is ignored
4. The requesting router then sends a triggered update out all interfaces that contains its routing table

IP address classes & classful routing



- ▶ **RIPv1** is a classful routing protocol – and therefore does not send subnet mask information in the update
- ▶ The router determines the subnet mask
 - Uses the subnet mask configured on a local interface
 - Or applies the default, classful subnet mask
- ▶ Because of this limitation, RIPv1 networks cannot be separated, nor can they implement VLSM

RIPv2 Protocol



- ▶ **RIPv2 (RFC 2453)** is compatible with RIPv1
- ▶ Subnet Mask applied on corresponding IP
- ▶ Routing at Hops: Next Hop IP Address tells which destination will be sent to the receiver routing table
 - If this field value is '0', the RIP message sender's address is considered (use of multiple protocols on, a single router)

RIPv2 benefits/downsides

► Benefits

- Widespread use and implemented on every routing equipment
- Handle subnet (Classless)
- Handle router authentication (MD5, uses Authentication keys – password)
- User friendly

► Downsides

- Metric (hop count)
- Limited to 15 hops
- Slow convergence (the ‘state’ of a set of routers that have the same topological information)

Display RIP Configuration

- ▶ Display information about the routing protocol used on each interface

```
epita#show ip protocols
```

- ▶ Display RIP routes:

```
epita#show ip route rip
      165.132.0.0/16 is variably subnetted, 3 subnets, 3 masks
R      192.168.2.0/24 [120/1] via 192.168.1.2, 00:00:25, Serial2/0
```

RIPv2 Configuration

- ▶ By default the router sends RIPv1 and can receive both versions

- Config:

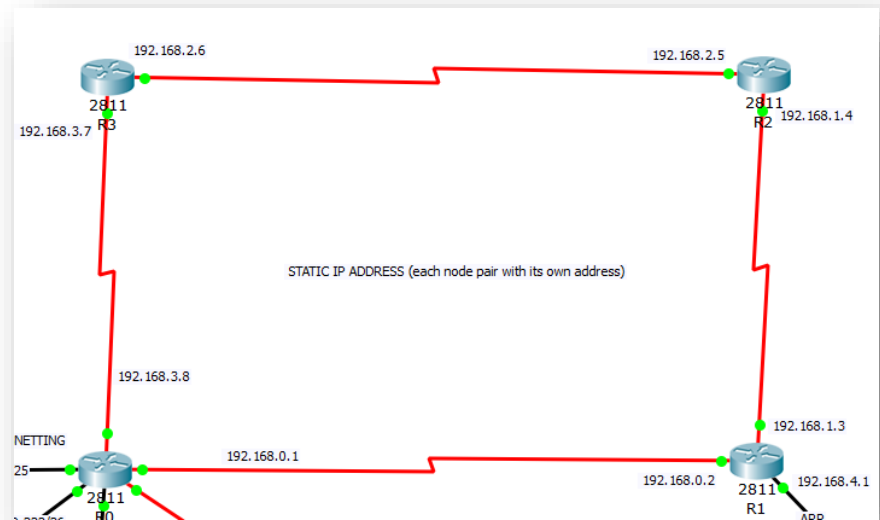
1. `Conf t`
2. `Router rip`
3. `Version 2` //To define version RIPv2
4. `Network x.x.x.x`
5. `exit`

- ▶ To use RIP only on a specified interface:

```
epita(config-if)#ip rip send version 2  
epita(config-if)#ip rip receive version 2
```

Exercise 8: Practical work

- ▶ Use your existing cisco packet tracer file (from last class exercise):
 1. Implement RIPv2 on all interfaces of 4 routers
 2. Verify using PING and TRACERT/TRACEROUTE



Save cisco packet tracer file with your 'First_Last-name_Ex-3-8'

Deadline: See 'Teams' Assignment section

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OSPF (Open Shortest Path First)

- ▶ Most widely used Interior Gateway Protocol (IGP)
- ▶ Supports Classless Inter-Domain Routing (CIDR)
- ▶ Used by Internet Gateways & Routers
- ▶ Uses Link State Routing (LSR) algorithm
 - Can create a complete view/map or topology of the network by gathering information from all other routers
 - Find Shortest Path First (SPF) using Dijkstra's algorithm
 - Link-state routing protocols do not use periodic updates
After the network has converged, a link-state update is only sent when there is a change in the topology

OSPF Operations

Tree

- Routers collect LS (Link State) information, LSA
- Network connection map (Tree) is formed

SPT

- LS routing algorithm forms loop-free Shortest path tree (SPT) to all destination nodes (Dijkstra's algorithm)

Gateway

- Routers update their routing tables (OSPF db description)
- Process gets repeated upon every change detection (multicast: *HELLO procedure*)

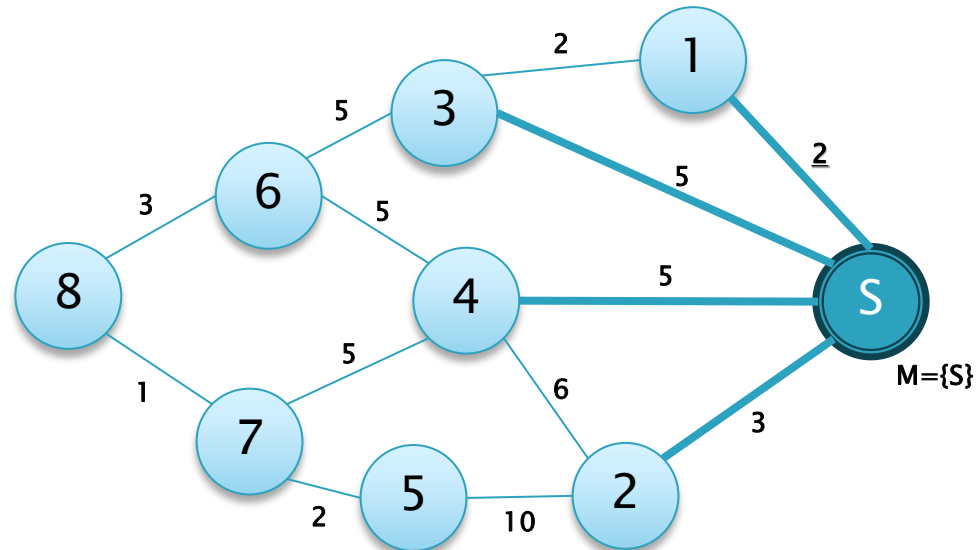
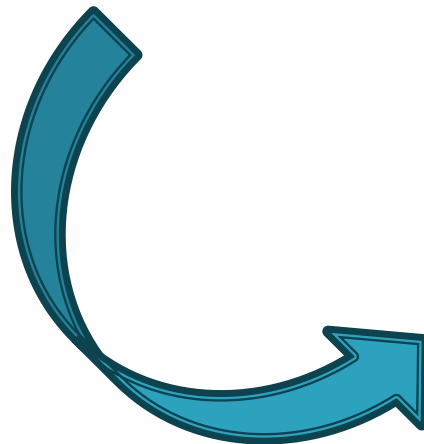
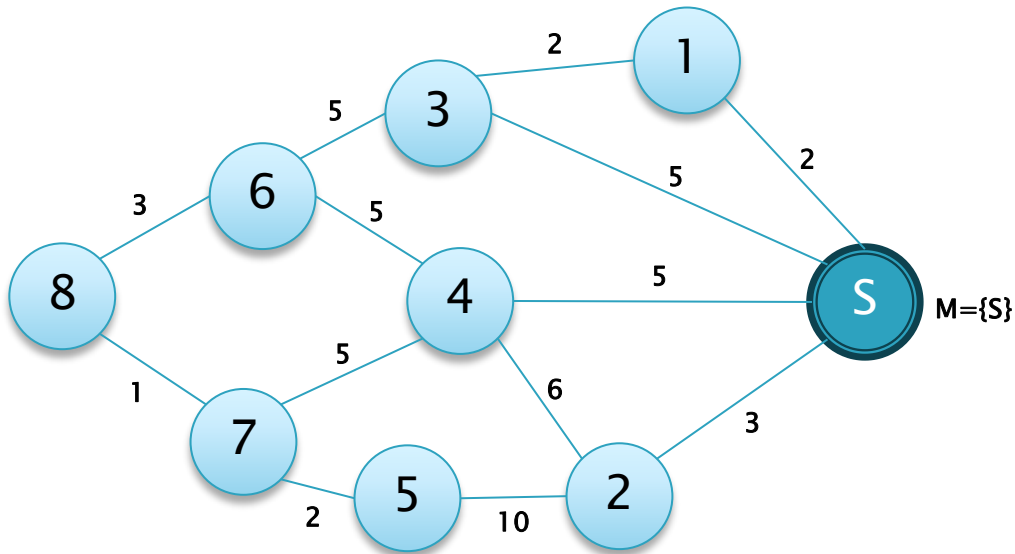
Link cost factors

- Throughput [bits/s, packets/s]
- Distance of a router
- RTT (Round-Trip Time) [s]
- Number of hops (routers/switches) to reach destination [hops]
- Availability [unitless]
- Reliability [unitless]
- ...

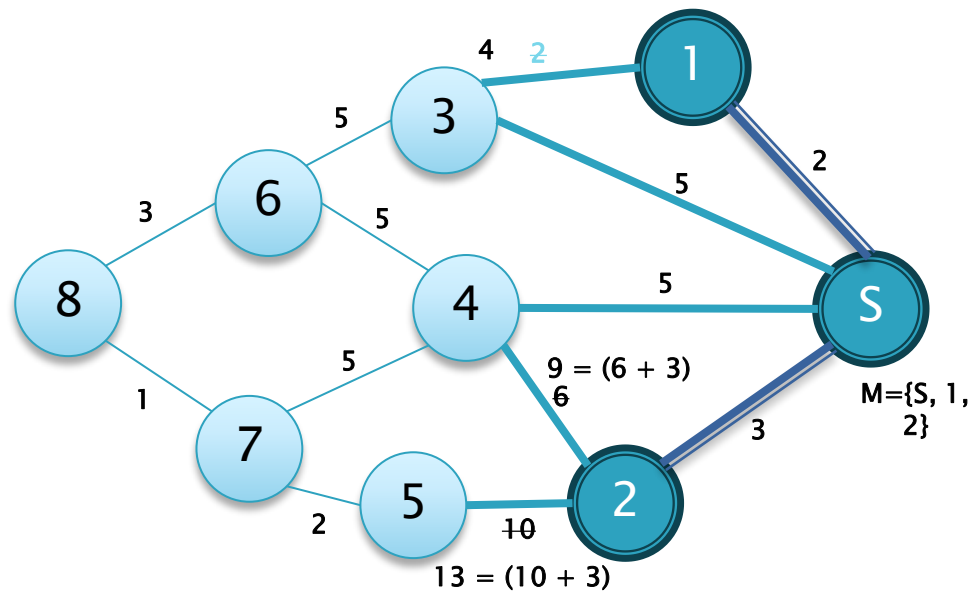
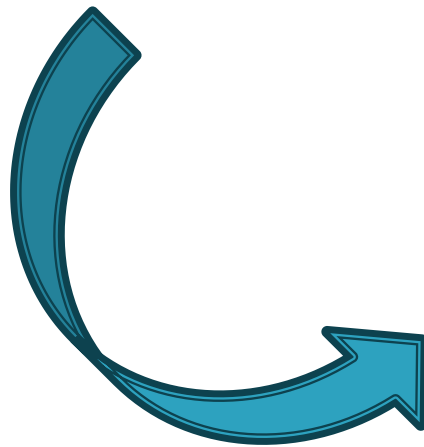
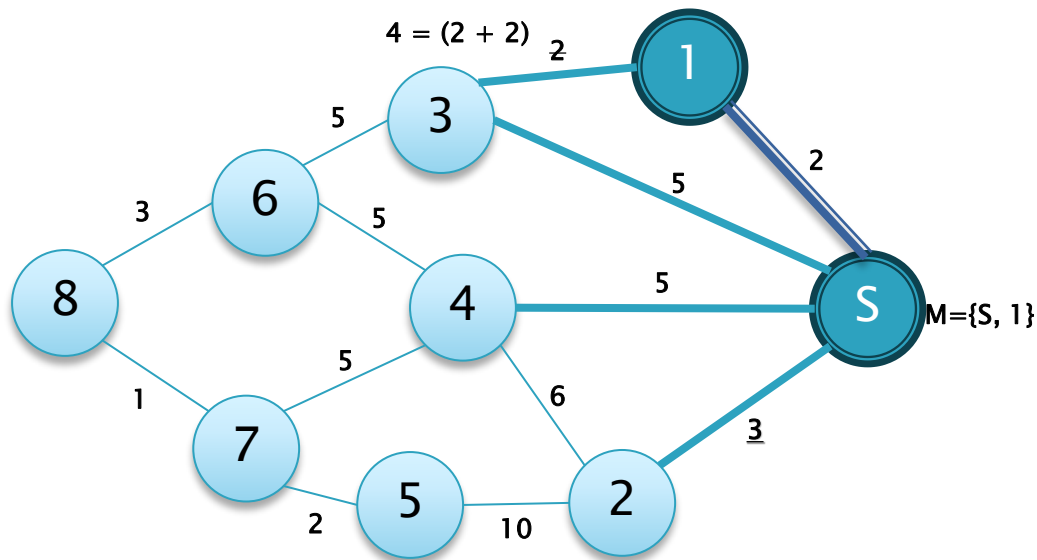
Shortest Path Algorithm (Dijkstra)

1. M is the set of nodes connected to the SPT
2. Initially M includes only the Source node (S)
3. Find the least cost route (by adding up link cost values on the routing path) from S to node X (that is not in set M yet), using nodes that exist in M as intermediate nodes
4. Add node X to set M
5. Record the least cost route from S to X
6. Repeat steps 3 to 5 until all nodes are in set M

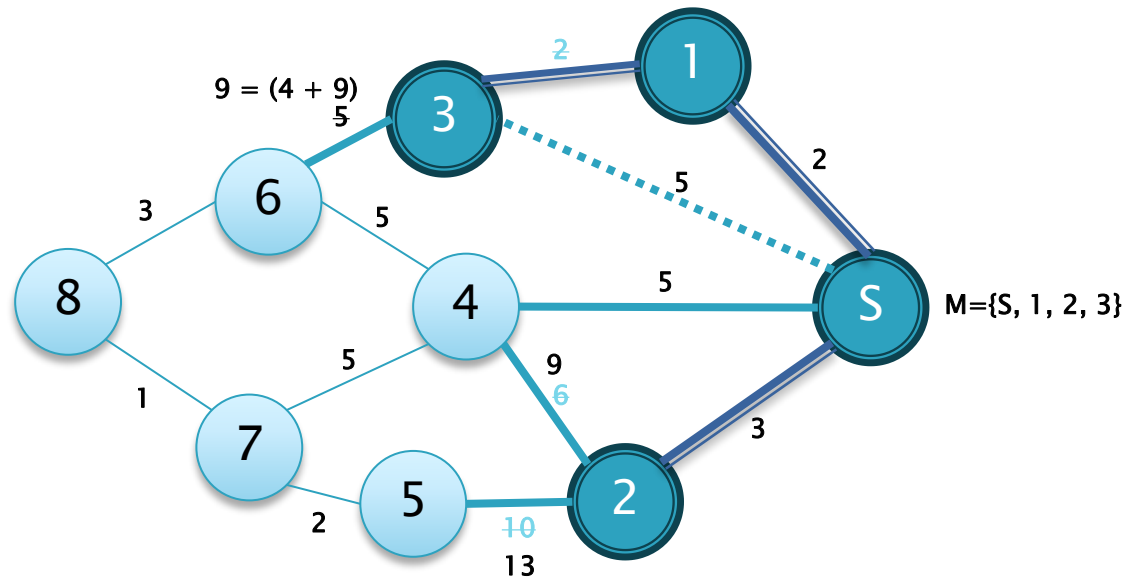
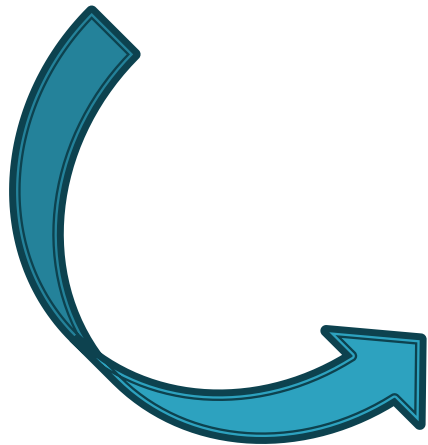
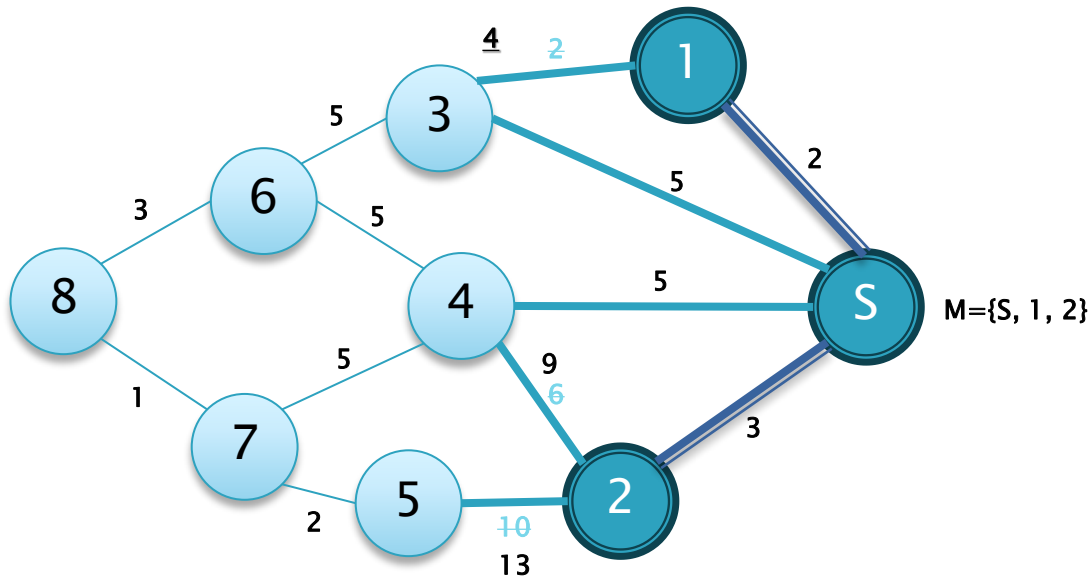
Shortest Path Algorithm (Dijkstra) – example



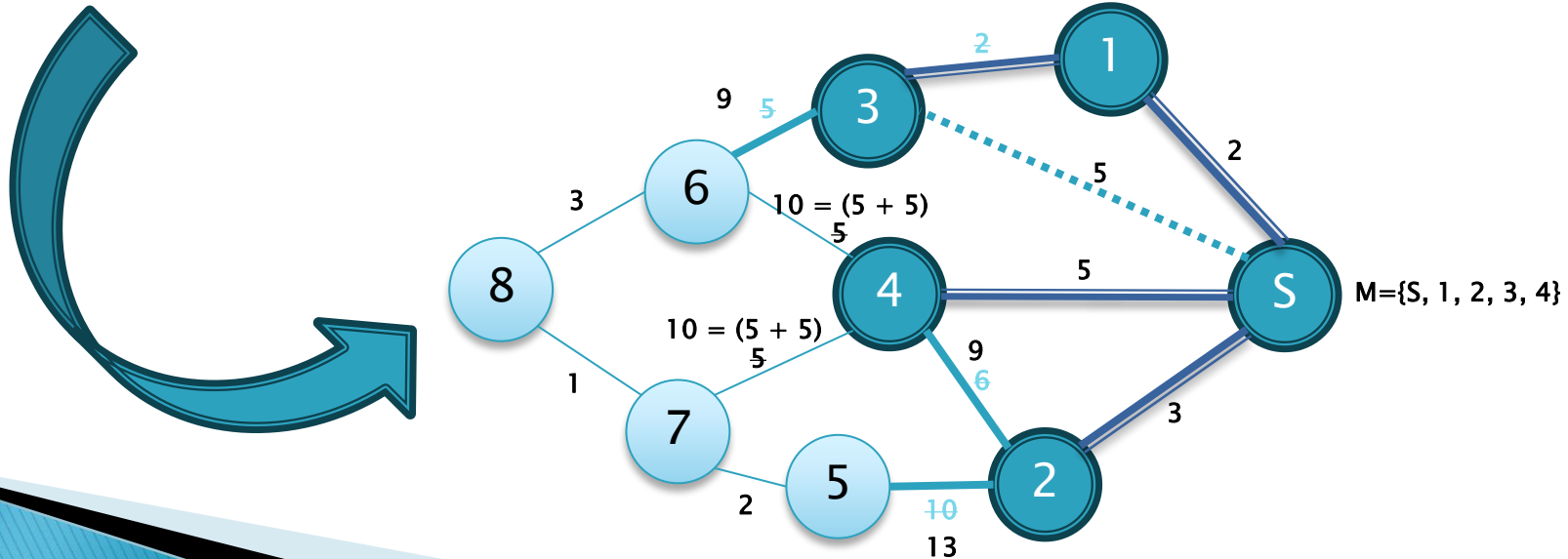
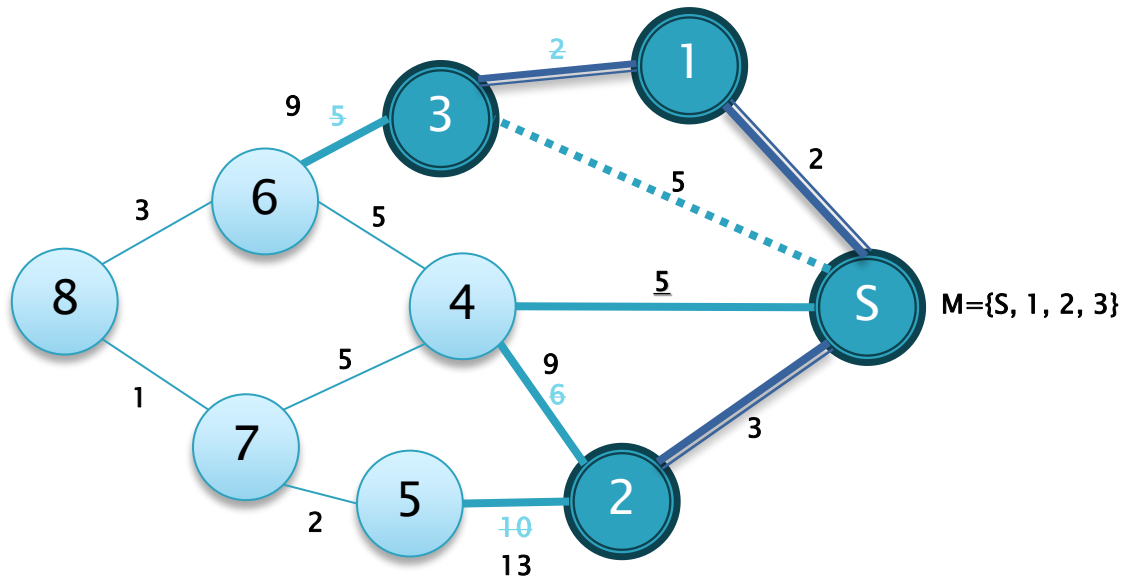
Shortest Path Algorithm (Dijkstra) – example (Cont.)



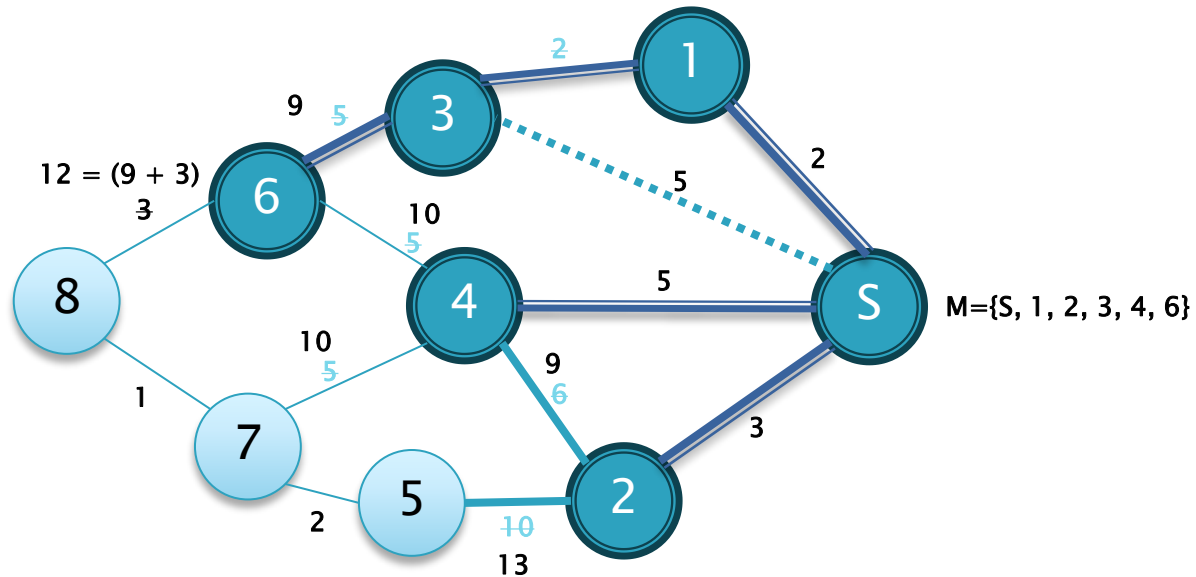
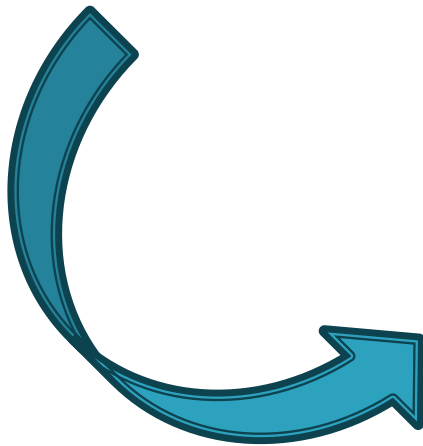
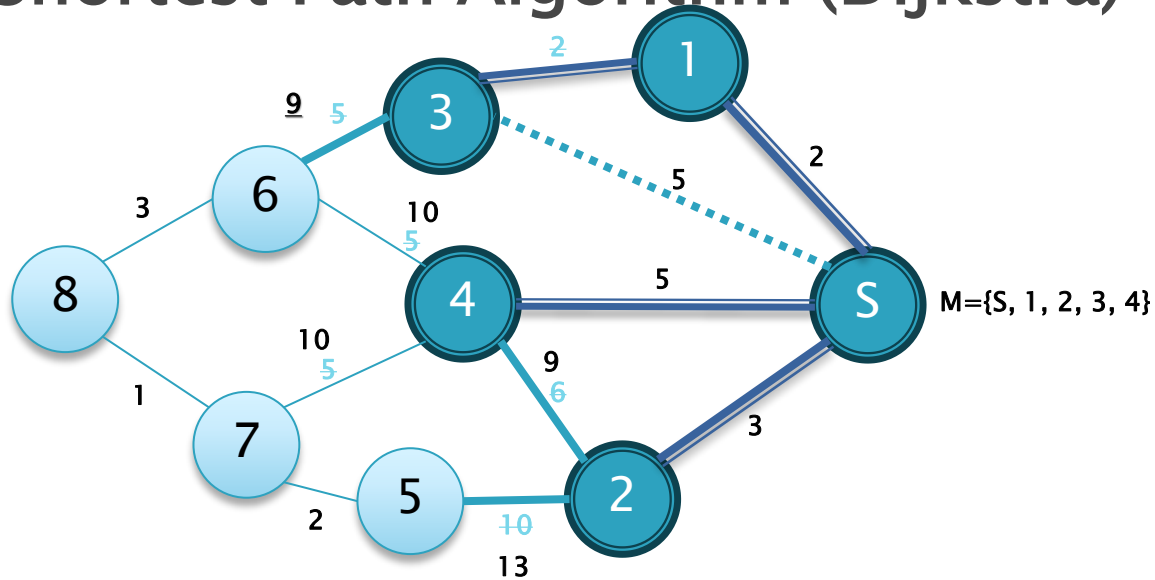
Shortest Path Algorithm (Dijkstra) – example (Cont.)



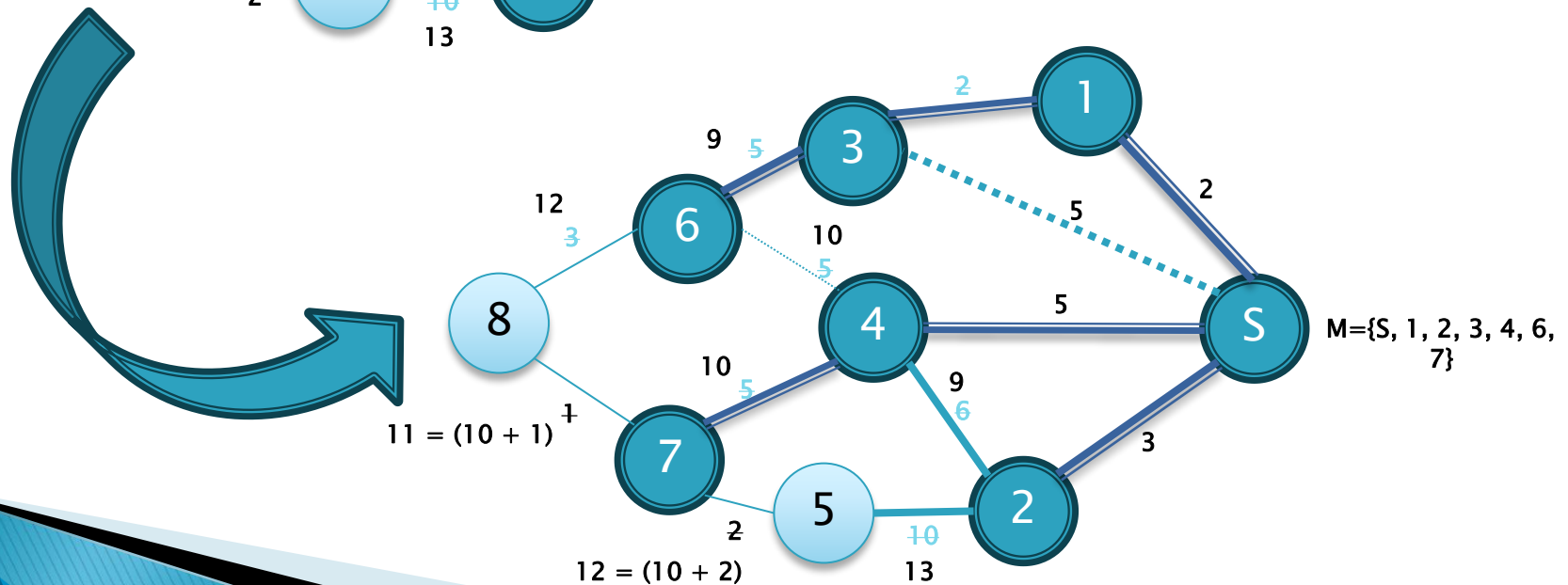
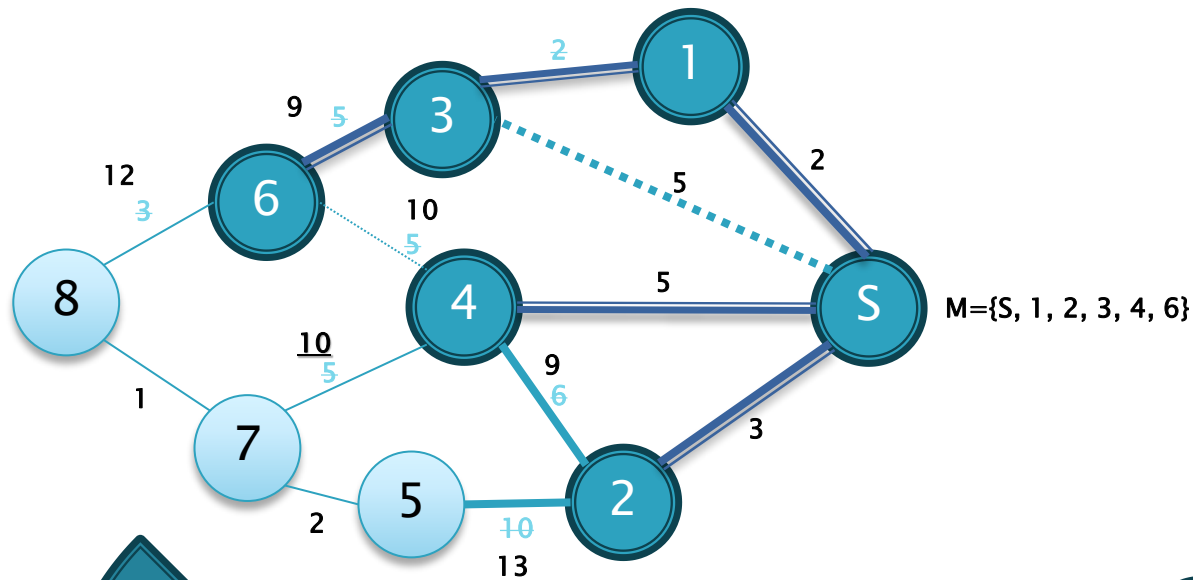
Shortest Path Algorithm (Dijkstra) – example (Cont.)



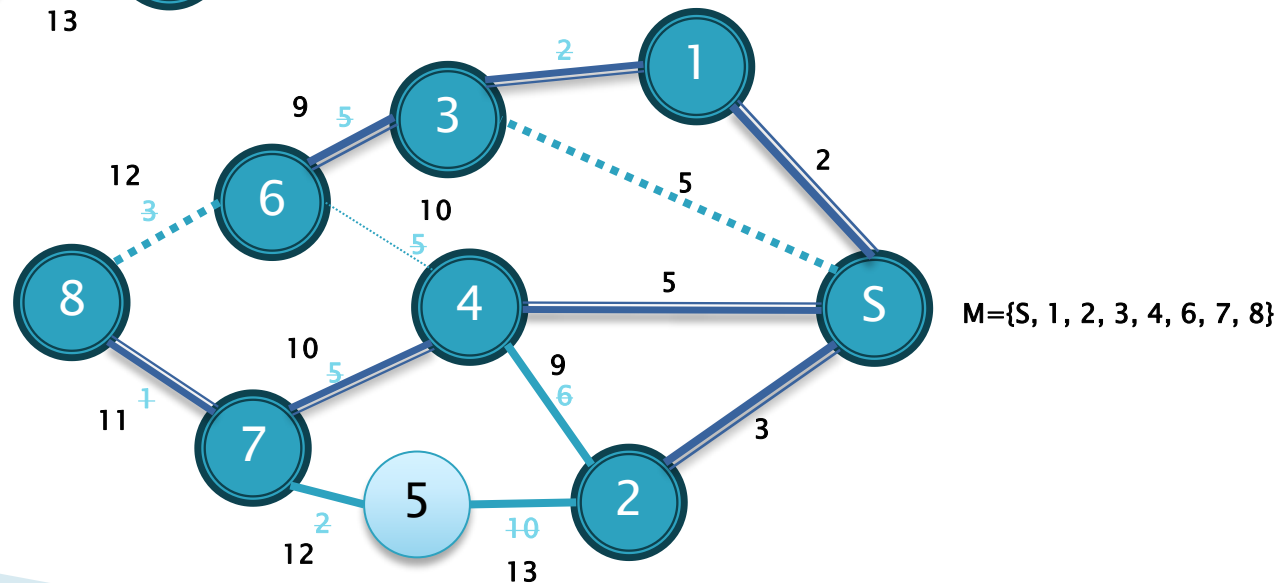
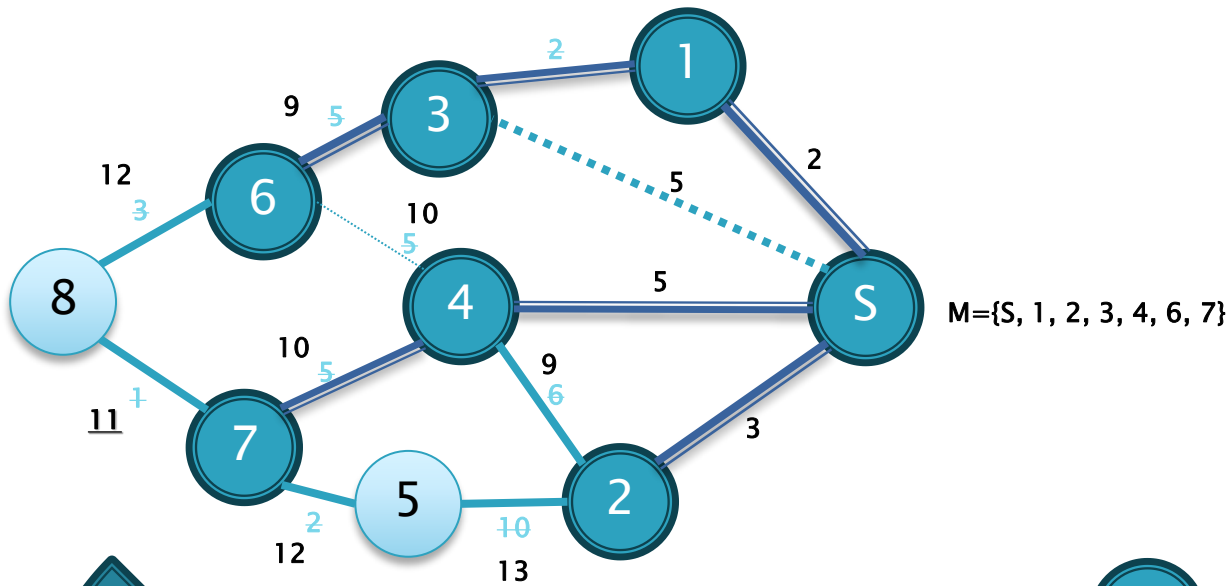
Shortest Path Algorithm (Dijkstra) – example (Cont.)



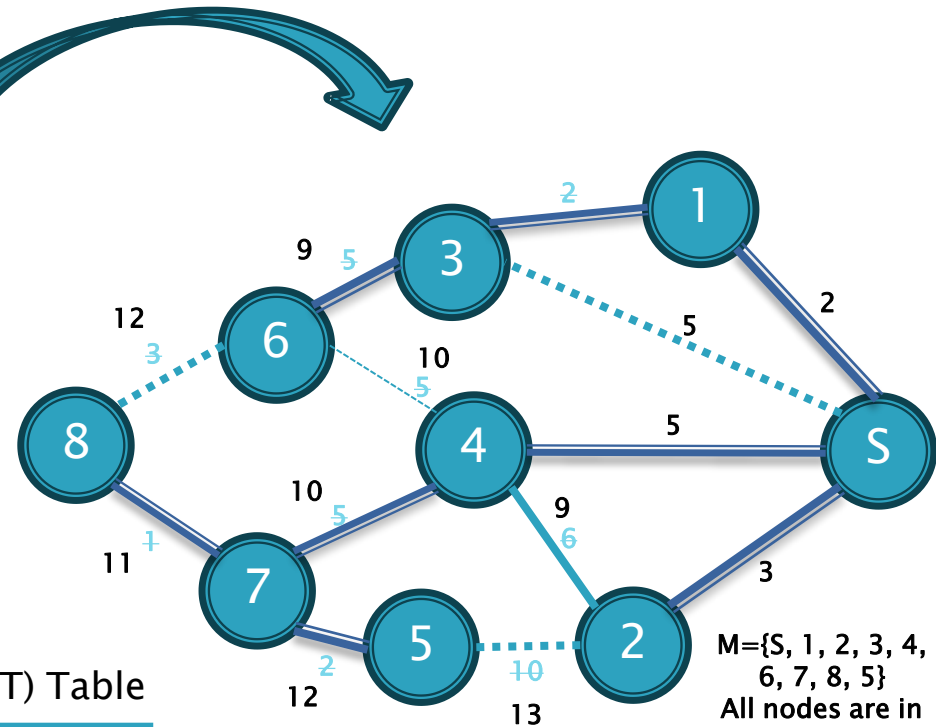
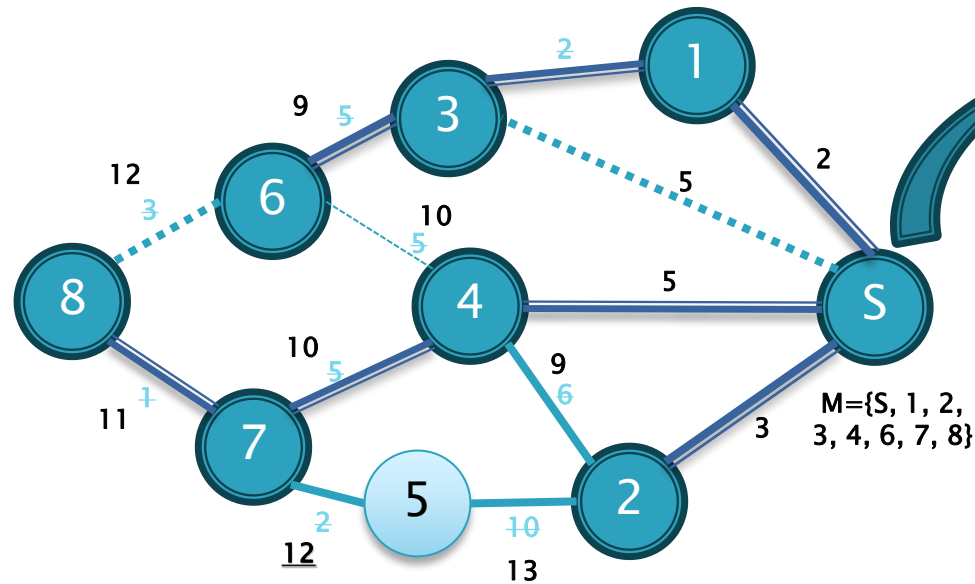
Shortest Path Algorithm (Dijkstra) – example (Cont.)



Shortest Path Algorithm (Dijkstra) – example (Cont.)



Shortest Path Algorithm (Dijkstra) – example (Cont.)



Shortest Path Tree (SPT) Table

From	To	Route
S	1	$S \rightarrow 1$
	2	$S \rightarrow 2$
	3	$S \rightarrow 1 \rightarrow 3$
	4	$S \rightarrow 4$
	5	$S \rightarrow 4 \rightarrow 7 \rightarrow 5$
	6	$S \rightarrow 1 \rightarrow 3 \rightarrow 6$
	7	$S \rightarrow 4 \rightarrow 7$
	8	$S \rightarrow 4 \rightarrow 7 \rightarrow 8$

OSPF subdivided areas

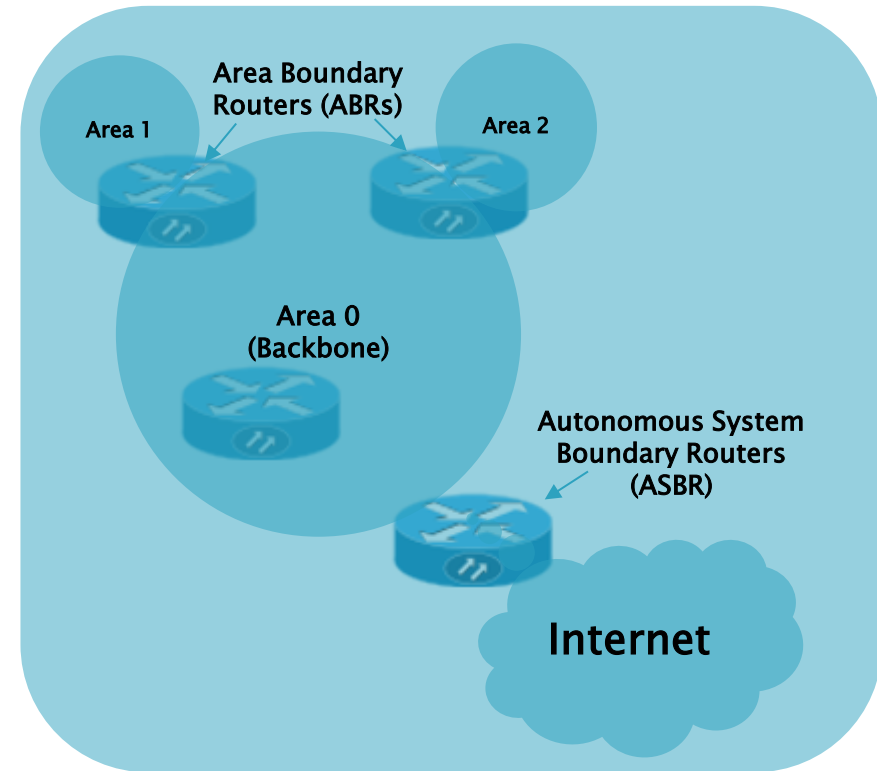
- ▶ An OSPF network can be structured, or subdivided, into routing **areas** (logical groupings of hosts and networks)
 - Division is carried out per Administration & Management requirements to:
 - Simplify administration (Traffic Engineering – TE)
 - Optimize traffic & resource utilization (Quality of service – QoS)
 - Enhance security
 - Enable faster routing updates
 - Areas are identified by 32-bit numbers (expressed in decimal/dot-decimal notation)
 - **Area 0 (zero)**, or 0.0.0.0, by convention represents the core or backbone area of an OSPF network
 - Additional areas must have a connection to the OSPF backbone area

i Always configure in order: **start by configuring Area 0 first**, then Area 1 and so on...

OSPF Router Types

- ▶ **IR (Internal Router):**
 - All routing interfaces belong to the same network area
- ▶ **ABR (Area Border Router):**
 - Connects subarea networks to the backbone network
 - Maintains separate link-state databases for each area it serves + summarized routes for all areas in the network
- ▶ **BR (Backbone Router):**
 - Connects to the backbone network
- ▶ **ASBR (Autonomous System Boundary Router):**
 - Connects between AS's using multiple routing protocols

Other Routers: DR (Designated router) and BDR (Backup designated router). DR has the responsibility to share 'Summary link states' with other areas. BDR serves for load-balancing



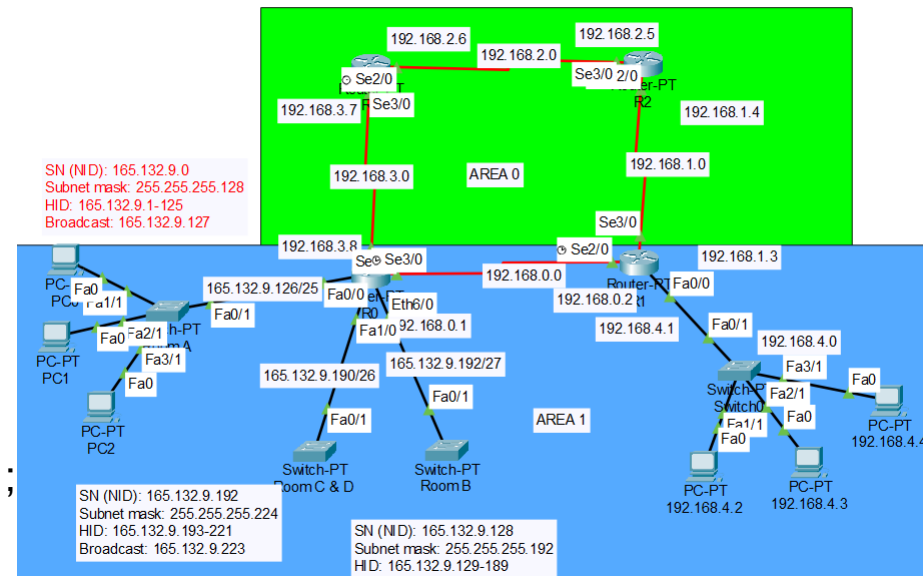
**Devices/routers
& Areas**

Exercise 9: Practical work

- Use your existing cisco packet tracer file (from last class exercise):

1. Remove RIPv2 on all interfaces
`Conf t`
`no router rip`
2. Configure OSPF:
Divide in 2 areas (Area 0: R4, R3;
Area 1: R0, R1):

```
conf t
router ospf 1 //1 is the process id
Network X.X.X.X wildcard (e.g.,
0.0.0.255) area 0 //wildcard is the
opposite of subnet mask
```



Verify using PING and
TRACERT/TRACEROUTE

*Save cisco packet tracer file with
your 'First_Last-name_Ex-3-9*

Deadline: See 'Teams' Assignment section

Lecture 3 ends here

- ▶ **Course Slides: Go to MS Teams:**
Introduction to Network protocols and architecture
→ Files section
- ▶ **Send your questions by email**
OR via direct message using MS Teams
- ▶ **Thank You!**